

DR. EVA C. HERBST

PEER-REVIEWED PUBLICATIONS * DENOTES CO FIRST AUTHOR, ** DENOTES STUDENT | SUPERVISED

1. | **Herbst EC**, Henninger HB, Stettler F**, Modelhart M**, Tanadini M, Ferguson SJ, Moroder P. 2026. Subject-specific glenohumeral translations in healthy human shoulders using biplanar fluoroscopy. *Accepted for Royal Society Open Science*.
2. | **Herbst EC**, van Beurden M**, Henninger HB, Ferguson SJ, Moroder P. 2026. Characterisation of variability and investigation of factors influencing maximum coronal abduction and forward elevation in post-operative rTSA subjects. *Multidisciplinary Biomechanics Journal: Proceedings of the International Shoulder Group Conference*.
3. | Shekhbihi A, Raiss P, Modelhart M, Randlkofer LM, **Herbst EC**, Moroder P. The Global Glenoid Component Inclination: why scapulothoracic orientation should be considered when defining glenoid component inclination in reverse total shoulder arthroplasty. *Journal of Shoulder and Elbow Surgery*.
4. | Shekhbihi A, Abdalla A, Modelhart M, **Herbst EC**, Raiss P, Moroder P. 2025. High variability in the degree of liner stability ratio among glenoid components of different anatomical total shoulder arthroplasty systems. *JSES International* 10(2).
5. | Modelhart M**, **Herbst EC**, Kunde AMH, Shekhbihi A, Hoffelner T, Scheibel M, Moroder P. 2025. While the bony glenoid is superiorly inclined, the glenoid surface is not. *JSES International*.
6. | Lappen S, Siegert P, Raiss P, Denard PJ, Werner BC, Griffin J, Shah A, Poltaretskyi S, Metcalfe N, **Herbst EC**, Moroder P. 2025. Influence of humeral liner constraint in reverse total shoulder arthroplasty on simulated range of motion in patients with various posture types – a modeling study. *The Hive Musculoskeletal Journal* 1(2).
7. | Moroder P, **Herbst EC**, Pawelke J, Lappen S, Schulz E. 2024. Large variability in the degree of constraint of reverse total shoulder arthroplasty liners between different implant systems. *Bone & Joint Open* 5(10).
8. | Akguen D, Gebauer H, Paksoy A, Schafer F, **Herbst EC**, Karczewski D, Pastor MF, Moroder P. 2024. Differences in Osseous Shoulder Morphology, Scapulothoracic Orientation, and Muscle Volume in Patients With Constitutional Static Posterior Shoulder Instability (Type C1) Compared With Healthy Controls. *Am. J. Sports Medicine* 52(5).
9. | Webb NM, Fornai C, Krenn VA, Watson LM, **Herbst EC**, Haeusler M. 2024. Gradual exacerbation of obstetric constraints during hominoid evolution implied by re-evaluation of cephalopelvic fit in chimpanzees. *Nature Ecology & Evolution* 8.
10. | Merten LJF, Manafzadeh AR, **Herbst EC**, Amson E, Tambusso PS, Arnold P, Nyakatura JA. 2023. The functional significance of aberrant cervical counts in sloths: insights from automated exhaustive analysis of cervical range of motion. *Proc. R. Soc. B.* 290 (2010)
11. | **Herbst EC***, Evans LA*, Felder AA, Javaheri B and Pitsillides AA. 2023. 3D profiling of mouse epiphyses across ages reveals new potential imaging biomarkers of early spontaneous osteoarthritis. *J. Anat.* 242(6)
12. | Demuth OE, **Herbst EC**, Polet DT, Wiseman ALA, Hutchinson JR. 2023. Modern three-dimensional digital methods for studying locomotor biomechanics in tetrapods. *J. Experimental Biol.* 226

13. | **Herbst EC**, Meade L, Lautenschlager S, Fioritti N, Scheyer TM. 2022. A toolbox for the retrodeformation and muscle reconstruction of fossil specimens in Blender. *Royal Society Open Science* 9(8)
14. | **Herbst EC**, Eberhard E, Manafzadeh AR, Richards C, Hutchinson JR. 2022. Multi-Joint analysis of pose viability supports the possibility of salamander-like hindlimb configurations in the Permian tetrapod *Eryops megacephalus*. *Integr. Comp. Biol.* 62(2).
15. | **Herbst EC**, Eberhard EA, Richards CT, Hutchinson JR. 2022. In vivo and ex vivo range of motion in the fire salamander *Salamandra salamandra*. *J. Anat.* 241(4)
16. | **Herbst EC***, Eberhard EA*, Hutchinson JR, Richards CT. 2022. Spherical frame projections for visualizing joint range of motion, and a complementary method to capture mobility data. *J. Anat.* 241(4)
17. | **Herbst EC**, Lautenschlager S, Bastiaans D, Miedema F, Scheyer TM. 2021. Modeling tooth enamel in FEA comparisons of skulls: comparing common simplifications with biologically realistic models. *iScience* 24(11)
18. | **Herbst EC**, Felder AA, Evans LAE, Ajami S, Javaheri B, Pitsillides AA. 2021. A new straightforward method for semi-automated segmentation of trabecular bone from cortical bone in diverse and challenging morphologies. *Royal Society Open Science* 8(8), figure featured on [journal cover](#).
19. | Ortega-Jimenez VM, **Herbst EC**, Leung MS, and Dudley R. 2020. Natural barriers: waterfall transit by small flying animals. *Royal Society Open Science* 7(8)
20. | **Herbst EC**, Doube M, Smithson TR, Clack J, and Hutchinson JR. 2019. Bony lesions in early tetrapods and the evolution of mineralized tissue repair. *Paleobiology* 45(4)
21. | **Herbst EC** and Hutchinson JR. 2018. New insights into the morphology of the Carboniferous tetrapod *Crassigyrinus scoticus* from CT. *Earth Environ. Sci. Trans. R. Soc. Edinb.* 109(1-2)

ABSTRACTS ACCEPTED FOR 2026 CONFERENCES

1. | **Herbst EC**, Lloyd JE, Moroder P, Ferguson SJ. *Automated Subject-Specific Modeling of 6 DoF Glenohumeral Kinematics*. Accepted for World Congress of Biomechanics, Vancouver, July 2026. (Podium Presentation).
2. | **Herbst EC**. *Subject-specific Shoulder Variability and Personalized Modeling for Orthopaedic Applications*. **Invited leading speaker** for Orthopaedic Biomechanics: Upper Extremities Track, World Congress of Biomechanics, Vancouver, July 2026.
3. | McCarthy L**, Heres J**, Ferguson SJ, Moroder P, Porto A, **Herbst EC**. *Statistical shape modeling for distal humerus reconstruction from routine clinical CT scans*. Accepted for Virtual Physiological Human (VPH), Milan. September 2026. (Poster Presentation)

SELECTED CONFERENCE PRESENTATIONS * DENOTES CO FIRST AUTHOR, ** DENOTES STUDENT | SUPERVISED

1. | **Herbst EC**, van Beurden M, Henninger H, Ferguson SJ, Moroder P. *Characterisation of variability and investigation of factors influencing maximum reach in post-operative rTSA subjects*. International Shoulder Group of International Society of Biomechanics, Lyon, June 2026. (Podium Presentation)

2. | Modelhart M**, Agbanyim D**, Cester D, Moroder P, Ferguson SJ, **Herbst EC**. *Interne Knochendichtekalibrierung des trabekulären Glenoids: erhöhte Genauigkeit durch Altersstratifizierung der Dichtewerte im Referenzgewebe*. DVSE Kongress, Karlsruhe, 2026. (Poster Presentation)
3. | Shekhbihi A, Raiss P, Modelhart M**, Randlkofer LM, **Herbst EC**, Moroder P. *Der Einfluss der skapulothorakalen Orientierung auf die Inklination der glenoidalen Komponente in der inversen Schultertotalendoprothetik*. DVSE Kongress, Karlsruhe, 2026. (Poster Presentation, **Winner of Best Poster Award**)
4. | **Herbst EC**, Lloyd JE, Roth L**, Moroder P, Ferguson SJ. *Subject-specific shoulder modeling with glenohumeral translation, contact, and muscle-driven kinematics*. 20th International Symposium on Computer Simulation in Biomechanics (TGCS), Uppsala. 2025. (Podium Presentation)
5. | **Herbst EC**, Stettler F**, Henninger H, Modelhart M, Tanadini M, Ferguson SJ, Moroder P. *Bony parameters and posture show no significant effects on glenohumeral translations in healthy subjects*. European Soc. of Biomech., Zurich. 2025. (Podium Presentation)
6. | **Herbst EC**, Stettler F**, Henninger H, Modelhart M, Tanadini M, Ferguson SJ, Moroder P. *Gesunde Schultern zeigen ein stabiles Drehzentrum*. D-A-CH Verein für Schulter- und Ellenbogenchirurgie (DVSE), Zurich. 2025. (Podium Presentation)
7. | Modelhart M**, **Herbst EC**, Kunde AMH, Shekhbihi A, Scheibel M, Moroder P. *Der Unterschied zwischen der knöchernen und globalen Glenoidinklination im Schultergelenk*. D-A-CH Verein für Schulter- und Ellenbogenchirurgie (DVSE), Zurich. 2025. (Podium Presentation)
8. | Lappen S, Siegert P, Raiss P, Denard P, Werner B, Griffin J, Shah A, Poltaretskyi S, **Herbst EC**, Moroder P. *Influence of Posture Types and Scapulothoracic Orientation on the Location of Bony Impingement after Reverse Total Shoulder Arthroplasty*. D-A-CH Verein für Schulter- und Ellenbogenchirurgie (DVSE), Zurich. 2025. (Podium Presentation). **Winner of Perthes Prize**
9. | **Herbst EC**, Lloyd JE, Sagl B, Ferguson SJ, Moroder P. *Towards patient specific FEM shoulder modeling. Invited speaker to Special Session "Multiscale Modeling Using Combined Multibody and Finite Element Simulation"*. 18th International Symposium on Computer Methods in Biomechanics and Biomedical Engineering, Vancouver. 2024. (Podium Presentation)
10. | **Herbst EC**, Ferguson SJ, Moroder P. *Testing scapulothoracic ellipsoid joint implementations using kinematics from biplanar fluoroscopy*. Annual Meeting of the Swiss Society for Biomedical Engineering (SSBE), Winterthur. 2024. (Poster)
11. | **Herbst EC**, Ferguson SJ, Moroder P. *The effect of scapulothoracic ellipsoid joint parameters on marker tracking errors*. European Soc. of Biomech., Edinburgh. 2024. (Podium Presentation)
12. | Agbanyim D**, Cester D., Ferguson SJ, Moroder P, **Herbst EC**. *Developing and validating a phantomless calibration method for BMD analysis of the glenoid*. European Soc. of Biomech., Edinburgh. 2024. (Podium Presentation)
13. | Stettler F**, Henninger H, Ferguson SJ, Moroder P, **Herbst EC**. *In vivo glenohumeral translations during arm elevation in healthy subjects*. European Soc. of Biomech., Edinburgh. 2024. (Podium Presentation)
14. | **Herbst EC**, Agbanyim D**, Cester D., Ferguson SJ, Moroder P. *Bestimmung der Knochendichte des Glenoids mit einer neuen phantomlosen CT-Kalibrierungsmethode*. D-A-CH Verein für Schulter- und Ellenbogenchirurgie (DVSE), Dusseldorf. 2024. (Podium Presentation)

15. Moroder P, **Herbst EC**, Pawelke J, Lappen S. *Large Variations In The Degree Of Constraint Of Reverse Total Shoulder Arthroplasty Liners Between Different Implant Systems*. SECEC (European Society for Surgery of the Shoulder and Elbow), Munich. 2024. (Podium Presentation). Abstract published in [JSES International](#) pg. P1341.
16. Bastiaans D, **Herbst EC**, van de Kamp T, Zuber M, Scheyer TM. *The first 3D cranial and myological reconstruction of the highly flattened remains of Askeptosaurus italicus (Diapsida: Thalattosauriformes)*. International Society for Vertebrate Morphology. Cairns, Australia. 2023. (Poster)
17. Evans LAE*, **Herbst EC***, Felder AA, Ajami S, Javaheri B, Pitsillides AA. *Do age-related differences in healthy and osteoarthritic mouse tibias show future imaging biomarkers?* Anatomical Society Summer Meeting, Glasgow. Abstract published in [Journal of Anatomy](#). 2021. (Podium Presentation)
18. **Herbst EC**, Lautenschlager S, Fioritti N, Meade L, Scheyer TM. 2021. *Modelling muscle volumes for finite element analysis and multibody dynamics*. XVIII International Symposium on Computer Simulation in Biomechanics. Online. 2021. (Podium Presentation)
19. Webb NM, Fornai C, Krenn VA, **Herbst EC**, Haeusler M. 2021. *A tight squeeze for chimpanzees: the role of joint laxity and fetal head orientation during birth*. European Society for the Study of Human Evolution. Online. Abstract published in [PaleoAnthropology](#), pg. 270. 2021. (Podium Presentation)
20. Evans LAE*, **Herbst EC***, Felder AA, Ajami S, Javaheri B, Pitsillides AA. *Do 3D epiphyseal bone architectural changes in ageing STR/Ort and healthy mice reveal early imaging biomarkers of osteoarthritis?* Bone Research Society Annual Meeting. Online. 2021. (Podium Presentation.) **Winner of New Investigator Award**. Abstract published in [JMBR Plus](#), pg. 6
21. **Herbst EC**, Eberhard EA, Manafzadeh AR, Richards CT, Hutchinson JR. *New methods support the possibility of a salamander-like walk in the Permian tetrapod Eryops*. Society for Integrative and Comparative Biology Annual Meeting, Online. 2021. (Podium Presentation). **Winner of D. Dwight Davis Award**
22. **Herbst EC**, Bastiaans D, Miedema F, Scheyer TM, Lautenschlager S. 2021. *How important is modeling tooth enamel in FEA comparisons of whole skulls? Comparing common simplifications with biologically realistic models*. Society for Integrative and Comparative Biology Annual Meeting, Online. 2021. (Poster)
23. Bastiaans D, **Herbst EC**, Scheyer TM. *Bringing fossils back to life: 3D cranial reconstructions of the highly flattened remains of thalattosauriformes*. Society for Integrative and Comparative Biology Annual Meeting, Online. 2021. (Podium Presentation)
24. **Herbst EC**, Eberhard EA, Manafzadeh AR, Richards CT, Hutchinson JR. 2020. *Was the early tetrapod Eryops capable of a salamander-like walk? Developing new methods to test paleontological hypotheses about posture and gait*. Swiss Geosciences Meeting, Online. 2020. (Podium Presentation). **Winner of Swiss Commission of Palaeontology Prize**
25. **Herbst EC**, Felder AA, Evans LAE, Javaheri B, Ajami S, Pitsillides AA. *A new automated method of segmenting trabecular bone: investigating subchondral trabecular changes as a predictor of osteoarthritis at the joint surface*. Bone Research Society Annual Meeting, Online. 2020. (Poster). Abstract published in [JMBR Plus](#) pg. 51
26. **Herbst EC**, Eberhard EA, Richards CT, Hutchinson JR. *Comparing in vivo and ex vivo knee range of motion in salamanders: a new method for investigating joint mobility*. CAMS-Knee OpenSim Workshop, ETH, Zürich. 2020. (Poster)

27. Bastiaans D, **Herbst EC**, Scheyer TM. *Re-fleshing fossils: cranial reconstructions of thalattosauriformes*. Swiss Geosciences Meeting, Online. 2020. (Podium Presentation)
28. Bastiaans D, **Herbst EC**, Webb NM, Haeusler M, Scheyer TM. *3D Data, a gateway to open science: the FEZ initiative*. OILS (Open Innovation in Life Sciences), Online. 2020. (Podium Presentation)
29. Bastiaans D, **Herbst EC**, Scheyer TM. *Virtual paleontology: a modern look at ancient material*. OILS (Open Innovation in Life Sciences), Online. 2020. (Podium Presentation)
30. Bastiaans D, **Herbst EC**, Scheyer TM. *Thalattosauriformes: schedelreconstructies van Triasische weirdos*. NKVP (Nederlandse Kring van Vertebraten Paleontologen), Online. 2020. (Podium Presentation)
32. **Herbst EC**, Eberhard EA, Richards CT, Hutchinson JR. 2019. *A new method for investigating joint mobility and its relevance for inferring locomotor evolution in early tetrapods*. 12th International Congress of Vertebrate Morphology, Prague, Czech Republic, abstract [here](#). 2019. (Podium Presentation)
32. **Herbst EC**, Doube M, Smithson TR, Clack J, Hutchinson JR. *Paleopathologies in Carboniferous tetrapods and the evolution of bone healing*. Society of Vertebrate Paleontology, 78th Annual Meeting, Albuquerque, New Mexico. 2018. (Poster)
33. CM Holliday, **Herbst EC**, M Jacoby, A Smolinsky, K Sellers. *Morphometric and modeling approaches to understanding the evolution of pseudosuchian mandibular symphyses*. Society of Vertebrate Paleontology, 78th Annual Meeting, Albuquerque, New Mexico. 2018. (Podium Presentation)
34. **Herbst EC** 2018. *New elements discovered in the early tetrapod Crassigyrinus scoticus*. DVM SICB Regional Meeting, Natural History Museum, London. 2018. (Podium Presentation)
35. **Herbst EC**, Smithson TR, Clack J, Doube M, Hutchinson JR. *Bony lesions in early tetrapods and the evolution of bone healing*. Society of Integrative and Comparative Biology Annual Meeting, San Francisco. 2018. (Podium Presentation)
36. **Herbst EC** and Hutchinson JR. *New insights into the morphology of the Carboniferous tetrapod Crassigyrinus scoticus gleaned from computer tomography*. The Early Tetrapod World Conference. University of Cambridge. 2017. (Invited Conference Podium Presentation)
37. **Herbst EC**, Smithson TR, Clack J, Hutchinson JR 2017. *Pathology in the early tetrapod Crassigyrinus scoticus*. Progressive Palaeontology Annual Meeting, University of Leicester. 2017. (Podium Presentation)

INVITED PRESENTATIONS AND WORKSHOPS

1. *In vivo and in silico subject specific variability in glenohumeral biomechanics*. Mathematical Laws of Morphology and Biomechanics Seminar Series (Online). 2025.
2. *Quantifying glenohumeral translations: how stable is the healthy shoulder joint, and how much variability exists between subjects?* International Women in Biomechanics Group (Online). 2025.
3. *Workshop: Building combined multibody/FEM Models using ArtiSynth*. Demonstration and hands-on workshop of Artisynth toolkit, with John Lloyd. TGCS Conference, Uppsala, Sweden. 2025.
4. *Glenohumeral Translations for Computational Models and Implant Design*. Guest Lecture and Dataset Sharing, Applied Biostatistics, ETH Zurich. 2024.

5. | *Computational tools to investigate 3D form and function in extinct and extant taxa*. Palaeontology Discussion Group Seminar Series, University of Bristol, UK. 2021.
6. | *Reconstructing feeding function in Triassic reptiles: computational methods for biomechanical analyses*. Public Colloquium Series, Palaeontological Institute and Museum, University of Zurich, Switzerland. 2021.
7. | *Workshop: Trabecular bone segmentation*. Senckenberg Museum and Research Institute, Frankfurt, Germany. [Recording available on Youtube](#). 2021.
8. | *Motion capture and computational approaches to investigate joint range of motion* Palaeontology Discussion Group, University of Birmingham, UK. 2021.
9. | *Workshop: [How to clean 3D meshes in Blender](#)*. FunkyMUG (Functional Morphology Users Group). [Recording available on Youtube](#). 2021.
10. | *New methods support the possibility of a salamander-like walk in the Permian tetrapod Eryops*. Comparative Zoology Lab, Humboldt University Berlin & Natural History Museum Berlin, Germany. 2021.
11. | *Investigating joint range of motion in salamanders and early tetrapods*. Evolutionary Morphology and Biomechanics Group, University of Liverpool, UK. 2020.